

Numerical Solution of a Fredholm Integral Equation Using the Nyström Method

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1 Introduction

We aim to find a numerical solution for the Fredholm integral equation of the second kind:

$$y(x) = 1 - x + \frac{2}{\pi} \int_0^{\pi/2} \cos(x - z) y(z) dz$$

The exact analytical solution is provided for validation purposes:

$$y(x) = 1 - x + \frac{2(4 - \pi)}{\pi^2 - 4} [\pi \cos(x) + 2 \sin(x)]$$

2 Numerical Discretization

We divide the integration interval $[0, \pi/2]$ into N equal sub-intervals. The width of each interval is $\Delta x = \frac{\pi}{2N}$. We choose the midpoints of these intervals as our quadrature nodes:

$$x_n = \frac{(n - 1/2)\pi}{2N}, \quad n = 1, 2, \dots, N$$

By evaluating the integral equation at each node x_n and applying a rectangular (midpoint) rule for the integral, we obtain:

$$\begin{aligned} y(x_n) &= 1 - x_n + \frac{2}{\pi} \sum_{m=1}^N \int_{x_m - \Delta x/2}^{x_m + \Delta x/2} \cos(x_n - z) y(z) dz \\ &\approx 1 - x_n + \frac{2}{\pi} \sum_{m=1}^N \cos(x_n - x_m) y(x_m) \frac{\pi}{2N} \\ &= 1 - x_n + \frac{1}{N} \sum_{m=1}^N \cos(x_n - x_m) y(x_m) \end{aligned}$$

3 Vectorization and Solution

The discretized equations can be expressed in matrix-vector form. Let us define:

- Solution vector: $\mathbf{y} = [y(x_1), y(x_2), \dots, y(x_N)]^\top$
- Forcing vector: $\mathbf{f} = [1 - x_1, 1 - x_2, \dots, 1 - x_N]^\top$

- Kernel matrix: $\mathbf{K} \in \mathbb{R}^{N \times N}$ where $\mathbf{K}_{nm} = \cos(x_n - x_m)$

The system simplifies to the following linear equation:

$$\mathbf{y} = \mathbf{f} + \frac{1}{N} \mathbf{K} \mathbf{y}$$

Rearranging to solve for the unknown vector $\mathbf{y}$:

$$\left(\mathbf{I} - \frac{1}{N} \mathbf{K} \right) \mathbf{y} = \mathbf{f}$$

$$\mathbf{y} = \left(\mathbf{I} - \frac{1}{N} \mathbf{K} \right)^{-1} \mathbf{f}$$

where $\mathbf{I}$ is the $N \times N$ identity matrix.